

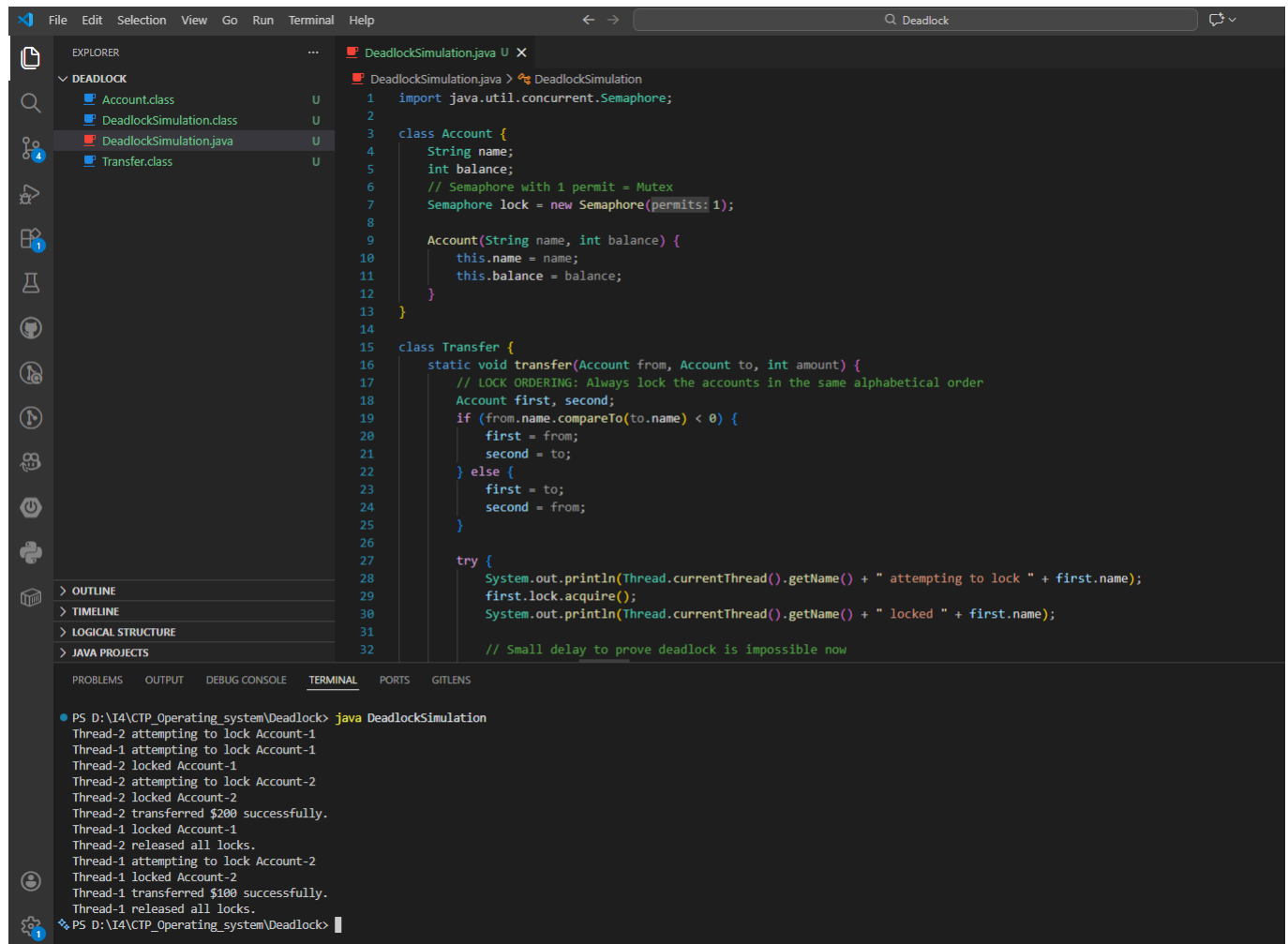
Operating System

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Deadlock



The screenshot displays an IDE with a project named 'DEADLOCK'. The Explorer view on the left shows the following files:

- Account.class
- DeadlockSimulation.class
- DeadlockSimulation.java
- Transfer.class

The main editor shows the source code for `DeadlockSimulation.java`:

```
1 import java.util.concurrent.Semaphore;
2
3 class Account {
4     String name;
5     int balance;
6     // Semaphore with 1 permit = Mutex
7     Semaphore lock = new Semaphore(permits: 1);
8
9     Account(String name, int balance) {
10         this.name = name;
11         this.balance = balance;
12     }
13 }
14
15 class Transfer {
16     static void transfer(Account from, Account to, int amount) {
17         // LOCK ORDERING: Always lock the accounts in the same alphabetical order
18         Account first, second;
19         if (from.name.compareTo(to.name) < 0) {
20             first = from;
21             second = to;
22         } else {
23             first = to;
24             second = from;
25         }
26
27         try {
28             System.out.println(Thread.currentThread().getName() + " attempting to lock " + first.name);
29             first.lock.acquire();
30             System.out.println(Thread.currentThread().getName() + " locked " + first.name);
31
32             // Small delay to prove deadlock is impossible now
```

The TERMINAL view at the bottom shows the output of running `java DeadlockSimulation`:

```
PS D:\I4\CTP_Operating_system\Deadlock> java DeadlockSimulation
Thread-2 attempting to lock Account-1
Thread-1 attempting to lock Account-1
Thread-2 locked Account-1
Thread-2 attempting to lock Account-2
Thread-2 locked Account-2
Thread-2 transferred $200 successfully.
Thread-1 locked Account-1
Thread-2 released all locks.
Thread-1 attempting to lock Account-2
Thread-1 locked Account-2
Thread-1 transferred $100 successfully.
Thread-1 released all locks.
PS D:\I4\CTP_Operating_system\Deadlock>
```

github link: <https://github.com/Nget-Darapich/Operating-System.git>